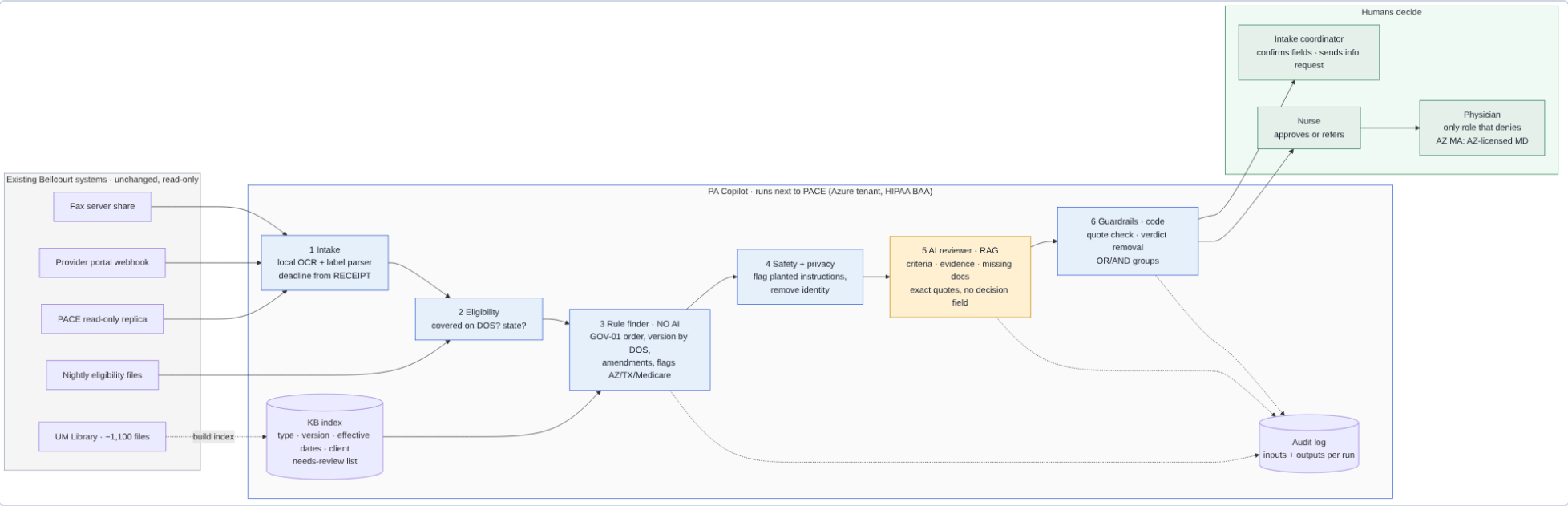
PA Copilot: Architecture

Components, data flow, where existing Bellcourt systems plug in, and where the human decides · Ratnesh Singhaniya · 5 October 2026



Blue = deterministic code · Amber = AI (one step) · Green = humans (all decisions) · Dotted = audit trail. Everything inside "PA Copilot" runs in Bellcourt's Azure tenant (HIPAA BAA) in production.

Data flow for one request

1. **Arrives** via the fax share, the portal webhook or the PACE replica. Intake reads it (faxes: local OCR + field labels) and starts the deadline from **receipt**.
2. **Eligibility**: covered on the date of service? Which state? Not covered → STOP to a human.
3. **Rule finder** (code): service + client + date → governing documents in GOV-01 order, the version in force on the date of service, amendments, memos marked "cannot change criteria", and flags (AZ sign-off, TX disclosure, Medicare criteria, PACE screen outdated). Unknown → STOP.
4. **Safety + privacy**: planted instructions flagged and removed; name, DOB, phone, MRN and member ID removed.
5. **AI reviewer**: selected sections + de-identified notes → criteria with exact quotes, evidence, plan provisions, missing documents. No decision field.
6. **Guardrails** (code): quotes and evidence verified, verdict sentences removed, OR-groups read whole.
7. **Humans**: the intake coordinator confirms fields and sends the information request; the nurse approves or refers; the physician decides any denial. The decision is recorded in PACE as today.
8. **Audit**: one record per run (inputs, documents sent, raw and checked answer, guardrail results). No identity.

Where existing systems plug in

System	Connection	Direction
PACE (no API, replacement 2028+)	Read-only database replica	Read only; never writes
Fax server	Watch the network share for new images	Read
Provider portal	Submission webhook calls PA Copilot	Read
Eligibility	Nightly client files	Read
UM Library (SharePoint)	Rebuild the index when files change	Read
Azure tenant (BAA)	Hosting, Azure OpenAI, Key Vault, logging	Runs here

Where the human decides

Decision	Who	Tool's role
Fields read from a fax are right	Intake coordinator	Shows fields, blanks and mismatches
Ask the provider for more information	Intake coordinator	Drafts the request with policy citations; sends nothing
Approve	Nurse	Criteria + evidence; no verdict
Deny (medical necessity)	Physician (AZ MA: AZ-licensed MD)	Flags who must sign; nothing else